PART 3: SUMMARY COMPARISON TABLE & CAMBRIDGE EXAM DRILL

Feature	Symmetric Encryption	Asymmetric Encryption
Number of Keys	1 key (shared secret key).	2 keys (mathematically linked Public & Private pair).
Encryption / Decryption	Same key encrypts and decrypts.	Public key encrypts; Private key decrypts.
Processing Speed	Very fast; low computational overhead.	Significantly slower (up to 1,000× slower).
Key Distribution	High risk: secret key must be shared across network.	Zero risk: public key can be shared openly anywhere.
Primary Usage	Encrypting large files, disk drives, live data streams.	Exchanging session keys, Digital Signatures, SSL handshakes.

Cambridge Exam Golden Tips

Tip 1 (Parity): When asked why a parity check failed to catch an error, always write: *"An even number of bits (e.g. 2 bits) flipped, so the parity count remained valid."*

Tip 2 (ARQ): Remember that an ARQ retransmission can be triggered by TWO things: either the sender receives a **NAK**, OR the sender's **timeout timer expires** before an ACK is received.

Tip 3 (Keys): In asymmetric encryption, remember the golden rule: *You ALWAYS encrypt using the RECIPIENT'S public key, and the recipient decrypts using their OWN private key.*